OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

15.3 Switches

Rail-bound vehicles use switches to change their tracks. In contrast to junctions, a switch can guide the vehicles into two directions only.

There are two different types of switches:

- Dynamic switches split the railroad track into two tracks leading in two directions. Dynamic switches can be changed during the simulation.
- Static switches split the railroad track into two tracks leading in two directions and have the two variants `straight` and `turn`. Static switches cannot be changed during the simulation.

Switches may be placed at an arbitrary position on a main track.

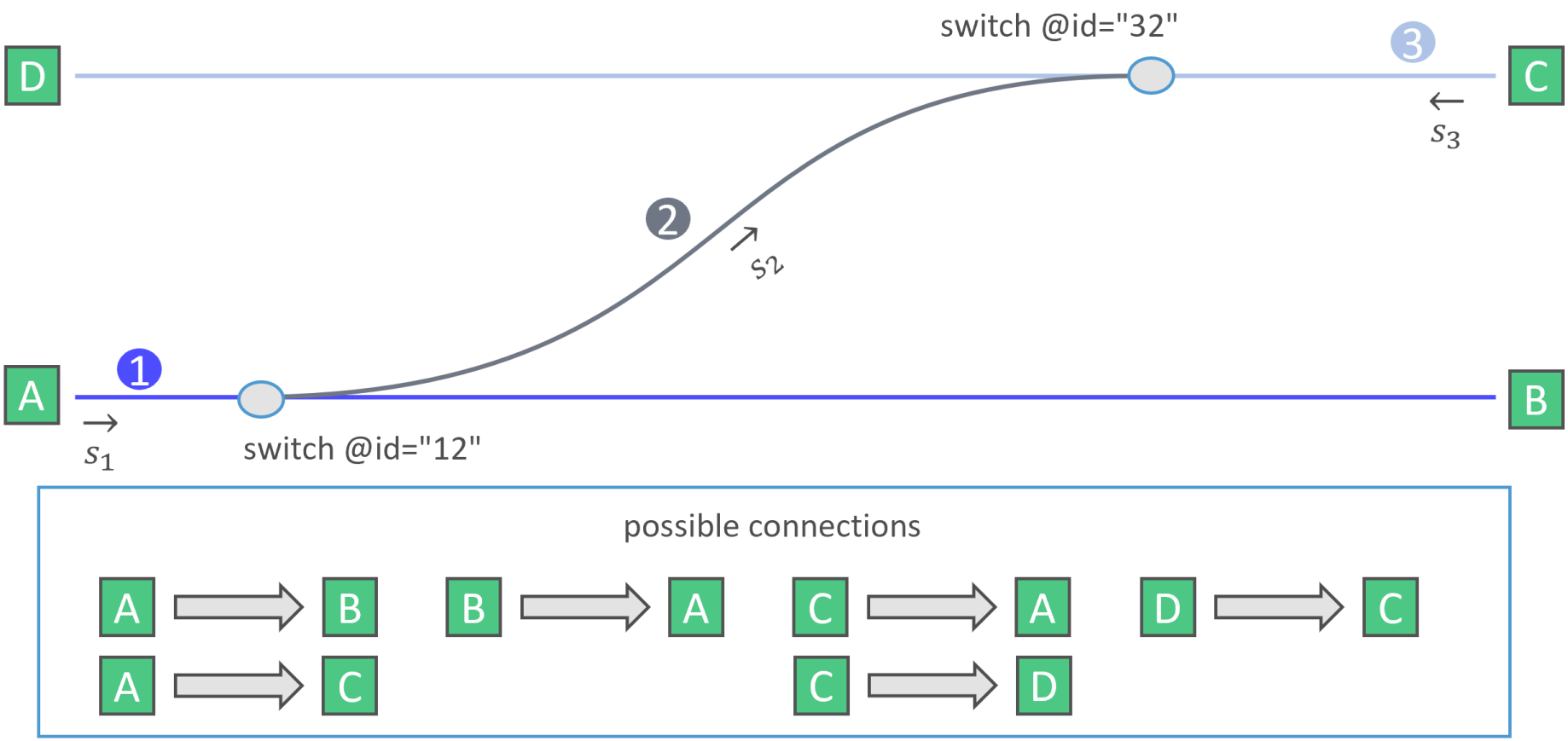

Figure 141. Railroad switches

Figure 141 shows the two partner switches `12` and `32`. A side track `2` connects the two main tracks `1` and `3`.

Elements in UML model

`<switch>` element

In ASAM OpenDRIVE, switches are represented by the `<switch>` element within the `<railroad>` element.

```
UML class: t_road_railroad_switch
XML tag:   <switch> (Multiplicity: 0..*)
```

Switches change the tracks for rail-bound vehicles. Switches guide the vehicles into two directions only.

Table 142. Attributes of the <switch> element

Name	Type	Use	Description
id	string	required	Unique ID of the switch; preferably an integer number, see uint32_t
name	string	required	Unique name of the switch
position	e_road_railroad_switch_position	required	Either a switch can be operated (dynamic) or it is in a static position

XML example

```
<railroad>
    <switch name="ExampleSwitch12" id="12" position="dynamic">
        <mainTrack id="1" s="1.0000000000000000e+01" dir="+"/>
        <sideTrack id="2" s="0.0000000000000000e+00" dir="+"/>
        <partner name="ExampleSwitch32" id="32"/>
    </switch>
</railroad>
```

Rules

The following rules apply to switches:

- A switch may be either dynamic or static.

Related topics

- Section 15.2, "Railroad tracks"
- Section 15.3.1, “Main track”
- Section 15.3.2, “Side track”

- Section 15.3.3, “Partner switches”

15.3.1 Main track

A main track represents the main course for rail bound traffic. A main track has the same properties as a side track. The two track types have been implemented as a convenience function to simplify the modeling of tracks entering and coming out of switches. Figure 141 shows a main track.

Elements in UML model

`<mainTrack>` element

In ASAM OpenDRIVE, main tracks are represented by the `<mainTrack>` element within the `<switch>` element.

```
UML class: t_road_railroad_switch_mainTrack
XML tag:   <mainTrack>
```

Main tracks form the primary course for rail bound traffic.

Table 143. Attributes of the <mainTrack> element

Name	Type	Use	Unit	Description
`dir`	e_elementDir	required		direction, relative to the s-direction, on the main track for entering the side track via the switch
`id`	string	required		Unique ID of the main track, that is, the `<road>` element. Must be consistent with parent containing this `<railroad>` element.
`s`	t_grEqZero	required	m	s-coordinate of the switch, that is, the point where main track and side track meet

Rules

The following rules apply to main tracks:

- asam.net:xodr:1.7.0:road.railroad.switch.check_switch_conn: Main tracks shall not be used to connect two switches.

Related topics

- Section 15.3.2, “Side track”
- Section 15.3.3, “Partner switches”

15.3.2 Side track

A side track connects two switches that are placed on main tracks. A side track has the same properties as a main track. The two track types have been implemented as convenience function to simplify the modeling of tracks entering and coming out of switches.

Figure 141 shows a side track.

Elements in UML model

`<sideTrack>` element

In ASAM OpenDRIVE, side tracks are represented by the `<sideTrack>` element within the `<switch>` element.

```
UML class: t_road_railroad_switch_sideTrack
XML tag:   <sideTrack>
```

Side tracks connect two switches that are placed on main tracks.

Table 144. Attributes of the <sideTrack> element

Name	Type	Use	Unit	Description

Name	Type	Use	Unit	Description
`dir`	e_elementDir	required		direction, relative to the s-direction, on the side track for after entering it via the switch
`id`	string	required		Unique ID of the side track, that is, the `<road>` element
`s`	t_grEqZero	required	m	s-coordinate of the switch on the side track

Rules

The following rules apply to side tracks:

- Side tracks shall be used to link two switches only.

Related topics

- Section 15.3.1, “Main track”
- Section 15.3.3, “Partner switches”

15.3.3 Partner switches

For convenience reasons, two switches may be declared partner switches. This describes a connection between two switches that are linked by a side track. These two switches need to be set consistently. Figure 141 shows the partner switches `12` and `32`.

Elements in UML model

`<partner>` element

In ASAM OpenDRIVE, partner switches are represented by the `<partner>` element within the `<switch>` element.

```
UML class: t_road_railroad_switch_partner
XML tag:   <partner> (Multiplicity: 0..1)
```

Partner switches are two consistently set switches linked by a side track.

Table 145. Attributes of the <partner> element

Name	Type	Use	Description
`id`	string	required	Unique ID of the partner switch
`name`	string	optional	Unique name of the partner switch

Rules

The following rules apply to partner switches:

- Partner switches shall be used to indicate that a side track links two switches.
- asam.net:xodr:1.7.0:road.railroad.switch.single_switch_no_partner: Single switches do not have partner switches.

Related topics

- Section 15.3.1, “Main track”
- Section 15.3.2, “Side track”